Introduction to Probability:

The discussion of probability starts with a random experiment. For our purposes, we limit ourselves to experiments that have a finite set of outcomes.

Some examples of random experiments suggested in class,

- Selecting a card out of a deck of 52 cards.
- Rolling a six-faced die.
- Tossing a coin.
- Observe nifty on any particular day.

Sample Space: For any random experiment, the set of all possible outcomes is called sample space.

Events: Any subset of the sample space is called an event.

Eg.

Consider the experiment of rolling a 6-faced die with labels $1, 2, \cdots, 6$ once.

Then, Sample Space $(S) = \{1, 2, 3, 4, 5, 6\}$

An example of an event is,

$$E = \{x : x \text{ is even}\}$$

which happens when the dice lands with an even numbered face.

Once we have learned about random experiments, sample space, and events, we can start making attempts to define "Probability" of an event. Intuitively, probability of an event indicates the "chances" of that event happening.

But how do we define it mathematically?

One approach is in terms of its 'relative frequency'. Suppose we repeatedly perform an experiment under same conditions. Let $n(E)$ denote the number of times event E occurs in the first n trials

Then probability of event E, denoted by $Pr(E)$, is defined as,

$$Pr(E) = \lim_{n \to \infty} \frac{n(E)}{n}$$

Now this definition, although it is very appealing, isn't a widely accepted definition for its own reasons.

Instead we have following three axioms that we work with,

Axioms of Probability:

1. for any event E,

$$0 \leq Pr(E) \leq 1$$

2. $Pr(S) = 1$, where S denotes the sample space.

3. Let $E_1, E_2, \ldots, E_n$ be mutually exclusive events, i.e., $E_i \cap E_j = \phi$, when $i \neq j$.

Then,

$$\Pr\left(\bigcup_{i=1}^{n} E_i\right) = \sum_{i=1}^{n} \Pr(E_i)$$

Some simple propositions :

Once we have agreed to the above 3 axioms, following propositions are easily derived,

1. $\Pr(\phi) = 0$

Proof: Since $S \cap \phi = \phi$, we have that,

$$\Pr(S \cup \phi) = \Pr(S) + \Pr(\phi) \quad \{\text{Axiom } 3.$$

$\therefore \quad \Pr(\phi) = 0$ $\blacksquare$

2. $\Pr(E^c) = 1 - \Pr(E)$

Proof: Again, $E \cap E^c = \phi$

So, $\Pr(E \cup E^c) = \Pr(E) + \Pr(E^c)$ $\{$ Axiom 3

Since $E \cup E^c = S$, we get,

$$Pr(S) = Pr(E) + Pr(E^c)$$

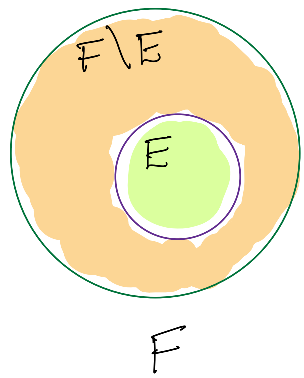
$$\therefore \quad Pr(E^c) = 1 - Pr(E) \qquad \left\{ \begin{array}{l} \text{Axiom 2,} \\ Pr(S) = 1 \end{array} \right.$$

$\boxed{\checkmark}$

3. If $E \subseteq F$, then $Pr(E) \leq Pr(F)$.

Proof: Observe that since $E \subseteq F$, we can write

$$F = E \cup (F \setminus E)$$
$$= E \cup (F \cap E^c)$$



Notice that $E \cap (F \cap E^c) = \phi$,

So, $Pr(E \cup (F \cap E^c)) = Pr(E) + Pr(F \cap E^c)$ {Axiom 3.

$$Pr(F) = Pr(E) + Pr(F \cap E^c)$$

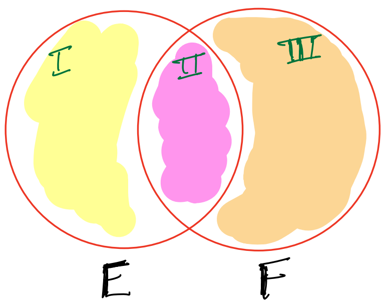
$$Pr(F) \geq Pr(E) \qquad \left\{ \begin{array}{l} \text{Since } Pr(F \cap E^c) \geq 0 \\ \text{by Axiom 1.} \end{array} \right.$$

4. For any two events E and F,

$$\Pr(E \cup F) = \Pr(E) + \Pr(F) - \Pr(E \cap F).$$

Proof: If events E and F are mutually disjoint, then the proposition follows immediately from Axiom 3 and proposition 1., i.e., $\Pr(\phi) = 0$.

So, if $E \cap F \neq \phi$. Then,



We have,

$$\Pr(E \cup F) = \Pr(\mathrm{I}) + \Pr(\mathrm{II}) + \Pr(\mathrm{III})$$

$$= \Pr(E \setminus (E \cap F)) + \Pr(E \cap F) + \Pr(F \setminus (E \cap F))$$

$$= \Pr(E \setminus (E \cap F)) + \Pr(E \cap F) + \Pr(F \setminus (E \cap F)) + \Pr(E \cap F) + \Pr(E \cap F) - \Pr(E \cap F) - \Pr(E \cap F)$$

Observe that $E\setminus(E\cap F)$ & $E\cap F$ are disjoint events and their union is E. Similarly we have for $F\setminus(E\cap F)$ & $(E\cap F)$.

So,

$$\Pr(E\cup F) = \Pr(E) + \Pr(F) - \Pr(E\cap F).$$

$\blacksquare$

Sample spaces having equally likely outcomes.

Let $S = \{1, 2, 3, \dots, N\}$.

Many often we assume that all the outcomes of the sample space are equally likely, i.e.,

$$\Pr(\{1\}) = \Pr(\{2\}) = \dots = \Pr(\{N\})$$

Let $\Pr(\{i\}) = p$, for $1 \leq i \leq N$.

Then, $\Pr(\{1\} \cup \{2\} \cup \dots \cup \{N\}) = \Pr(\{1\}) + \Pr(\{2\}) + \dots + \Pr(\{N\})$

Since all singleton events are disjoint.

So, $Pr(S) = p \cdot N$ {Because all outcomes are equally likely.

Therfore, $p = \frac{1}{N}$ {Axiom 2, $P(S) = 1$.

$\therefore$ $Pr(\{i\}) = \frac{1}{N}$, for $1 \leq i \leq N$.

An easy extension of this proof gives,

$$Pr(E) = \frac{|E|}{N},$$

when all outcomes are equally likely & $N = |S|$.

Birthday Paradox:

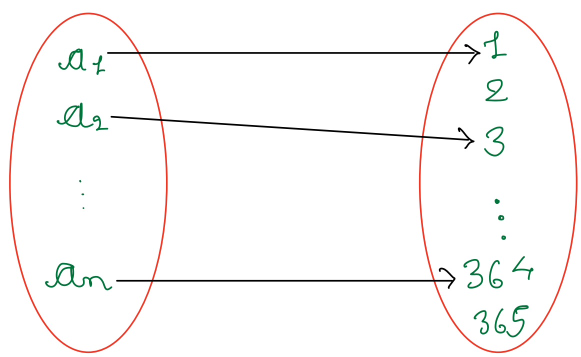
There are n people in a room. If we assume that each person is equally likely to be born on any of the 365 days of the year, what is the probability that no two of them have birthday on the same day? How large need n be so that this probability is less than $1/2$?

The size of the sample space is $(365)^n$.

Here is one way to see it,

Let $A = \{a_1, a_2, \ldots, a_n\}$, the set of n people and $B = \{1, 2, 3, \ldots, 365\}$ the set of days.

Now consider functions from A to B, that basically map people to their birthdays.



a_1 gets mapped to 1 means a_1 was born on the 1st day of the year and similarly for other a_i's.

Now every possible outcome is basically a function from A to B and in fact, every function corresponds to an outcome.

So counting total number of outcomes is equivalent to counting total no. of functions which we know is $(365)^n$.

Now the event that we are concerned about essentially corresponds to a one-to-one function from A to B. (Do you see why?).

We know that the number of one-to-one functions is, $(365)_n$.

Therefore, the required probability is

$$\frac{(365)_n}{(365)^n}.$$

for $n \geq 23$ this value is less than 1/2 which we'll try to verify programmatically.